

Stabilizing Goal-Conditioned PPO for Brawlhalla RL Agent

1 Goal Redesign

1.1 Goal Factorization

Introduce a discrete goal type:

$$g = (g_{\text{type}}, \tilde{g})$$

Core goal types:

- Recovery
- Approach
- Spacing
- Attack
- Weapon Acquisition

1.2 Goal Definitions

Recovery

$$g = (target_x, target_y, urgency)$$

Approach

$$g = (target_rel_distance, vertical_alignment, facing_required)$$

Spacing

$$g = (desired_distance, maintain_direction)$$

Attack

$$g = (in_range, attack_type, opponent_state)$$

Weapon

$$g = (weapon_dx, weapon_dy, pickup_flag)$$

1.3 Goal Masking

$$g_{\text{full}} = \text{mask}(g_{\text{type}}) \odot g$$

1.4 Remove Unstable Features

- Remove opponent damage from LLC goals
- Convert offstage into goal trigger
- Keep only relevant features per goal

1.5 Goal Sampling Distribution

$$P(g) = 0.4 G_{\text{core}} + 0.4 G_{\text{HER}} + 0.2 G_{\text{archive}}$$

1.6 Temporal Consistency

Goals persist for k steps:

$$g_t = g_{t+1} = \dots = g_{t+k}$$

2 Training Modifications

2.1 PPO with Anchoring

$$L = L_{\text{PPO}} + \lambda_{KL} \cdot D_{KL}(\pi \parallel \pi_{\text{old}}) + \lambda_{BC} \cdot L_{BC}$$

BC Loss on archive trajectories

2.2 Replay Integration

Training batch:

- 70% on-policy PPO
- 30% replay (KL + BC only)

2.3 Goal-Balanced Batching

Ensure equal sampling across goal types.

2.4 Advantage Normalization

$$A^{(g)} \leftarrow \frac{A^{(g)} - \mu_g}{\sigma_g}$$

2.5 Gradient Interference Mitigation

Use PCGrad or gradient projection to avoid destructive updates.

2.6 FiLM Regularization

Constrain modulation:

$$\gamma \in [0.5, 1.5], \quad \beta \in [-1, 1]$$

2.7 Curriculum Learning

Phase 1: Train each goal independently

Phase 2: Train pairs of goals.

Phase 3: Full mixed training

2.8 Policy Snapshot Anchoring

Maintain past policies:

- Store every N updates
- Keep last K snapshots
- Sample randomly for KL constraint

2.9 Replay Prioritization

Prioritize successful trajectories:

- Recovery success
- Successful hits
- Stable spacing

2.10 Evaluation Metrics

Track per goal type:

- Success rate
- Retention over time

3 Result

These modifications:

- Stabilize goal distribution
- Reduce skill interference
- Prevent catastrophic forgetting
- Prepare transition to hierarchical RL